

Multi-transformer based on prototypical enhancement network for few-shot relation classification with domain adaptation

Hongfei Du^{a,b,c,d}, Qinghua Zhang^{b,c,d,*}, Keyuan Li^{a,b,c,d}, Fan Zhao^{a,b,c,d}, Guoyin Wang^{b,c,d}

^a School of Computer Science and Technology, Chongqing University of Posts and Telecommunications, Chongqing, 400065, China

^b Key Laboratory of Big Data Intelligent Computing, Chongqing University of Posts and Telecommunications, Chongqing, 400065, China

^c Chongqing Key Laboratory of Computational Intelligence, Chongqing University of Posts and Telecommunications, Chongqing, 400065, China

^d Key Laboratory of Tourism Multisource Data Perception and Decision Ministry of Culture and Tourism, Chongqing University of Posts and Telecommunications, Chongqing, 400065, China

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ABSTRACT

Few-shot relation classification aims to predict the relation between entity pairs in unseen sentences with a few labeled sentences. In real-world scenarios, the target domains often have limited data and could differ significantly from the source domains. Existing studies focus on the representation of entity pairs, ignoring the relation between the sentences and the entity pairs. Some researchers use feature-adaptive methods that do not consider that features contribute differently to relation classification. To solve these challenges, a multi-transformer based on a prototypical enhancement network (MTPEnet) is presented for few-shot relation classification with domain adaptation in this study. Specifically, for MTPEnet, two main parts are conducted. First, in the calculation of the prototype, a local and global representations (LGR) module is established to improve the representation of sentences. LGR considers not only the head and tail entities in a sentence but also the relation between the entity pair and the entire sentence. Second, a multi-transformer (MT) feature transformation module is introduced to address the cross-domain problem. Different weights are assigned to the target domain features according to the degree of fusion between the features in the target and source domains. Finally, the experimental results on the datasets FewRel 2.0 and FewTAC demonstrate the superiority of MTPEnet over several baseline models.

1. Introduction

Relation classification (RC) aims to identify the semantic relations between the head and tail entities, which is a fundamental problem in NLP. It is used in many real-world applications such as information extraction [1], question-answering systems [2,3], and sentiment analysis [4]. Traditional RC models are trained using a large amount of labeled data. However, labeling data is labor-intensive, time-consuming, and prone to labeling errors, particularly in the professional field. Therefore, some models have been proposed using the distant supervision (DS) method [5,6] to alleviate this problem. The DS method is prone to inevitable noise problems because it assumes that the same pairs of entities have the same relation in the knowledge graphs. Considerable efforts have been devoted to noise reduction [7–9]. When dealing with long-tail relations for which only limited training data are available, the classification performance of the models decreases significantly.

To address the issue of long-tail in RC, Han et al. [10] proposed a few-shot relation classification (few-shot RC) dataset named FewRel and experimented with several basic few-shot learning models, including Meta Networks [11], GNN [12], SNAIL [13], and Prototypical Networks [14]. Prototypical networks have achieved good performance in few-shot learning. The few-shot RC task has become a research hotspot, and significant progress has been made with the introduction of the BERT pre-training language model [15]. In the realistic scenario, the cross-domain ability of the model becomes crucial as the test domains usually lack annotations and may be highly different from the source domains. Gao et al. [16] introduced the challenge of domain adaptation for few-shot RC and developed a new dataset FewRel 2.0, as shown in Table 1. The training and testing data originate from different domains, the former from Wikipedia and the latter from the medical corpus PubMed. In general, few-shot RC with domain adaptation faces two main challenges, i.e., (1) how to utilize limited data effectively, and (2) how to solve the problem that the model is trained on the source

* Corresponding author at: Key Laboratory of Big Data Intelligent Computing, Chongqing University of Posts and Telecommunications, Chongqing, 400065, China.

E-mail address: zhangqh@cqupt.edu.cn (Q. Zhang).

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Table 1

A case for few-shot relation classification dataset FewRel 2.0. Red indicates head entities, and blue indicates tail entities.

	Source domain		Target domain	
Support set	(A)director	She was the Acting Coach for the Oscar nominated short film Two Cars, One Night directed by Taika Waititi .	(A)ingredient_of	twenty-five patients were threatened with intravitreal bevacizumab injection 1 to 154 days prior to surgery (bevacizumab cnv).
	(B)country	Belgium's highest point is the signal de botrange at 694 meters above the sea level.	(B)occurs_in	three infants had congenital eyelid bands, ankyloblepharon filiforme adnatum (afa), in association with cleft lip and palate .
Query set	(A)or(B)?	The Ulster House Hotel, formerly the Wellington Hotel, is located on Main Street in Pine Hill , New York, United States .	(A)or(B)?	with advances in obstetric and perinatal management, the incidence of intraventricular hemorrhage in premature infants has declined, while periventricular leukomalacia remains a signification concern.

domain and then tested on the target domain, and the classification effect is degraded due to inconsistent data distribution.

The first challenge arises from the scarcity of training data for each relation. Fully exploiting available information is crucial because there is a limited amount of training data. Several studies [17,18] have incorporated relation information (such as relation names or interpretations) or integrated other knowledge bases to aid model learning. Although these methods yield promising results, they rely on additional knowledge bases that may not always be available in certain domains. Liu et al. [19] proposed a straightforward and efficient method for incorporating existing relation information into a model. However, previous studies have only comprehensively considered the relational representation and the characteristics of the entity pair in the semantic representation of the sentence. The sentence information is crucial for prototype computations. A good prototype is essential for accurately representing the relation and helping the model comprehend the semantics better. Inspired by the importance of good prototypes, a local and global representations (LGR) module is designed to embed sentences. The entity pair of the sentence and the entire sentence is considered, which helps us obtain a more informative and robust initial prototype for a better representation of the relation.

The second challenge refers to improving the domain shift caused by training data and testing data with different distributions. The mapping-based approach that maps the features of the source and target domains into a similar feature space is commonly used to solve domain-adaptive problems. Tang et al. [20] used affine functions to map features to learn a variety of different distributions and introduced an attention fusion mechanism to fuse extra information. Yin et al. [21] attempted to address the domain adaptation problem in few-shot RC using an attention mechanism designed to obtain domain-invariant clues. Although classification accuracy has improved, previous methods still have some limitations. In particular, the feature transformation does not fully consider the relation between the target and source domain data. It maps features from each domain independently. Li et al. [22] used the transformer to map features and demonstrated the effectiveness of the transformer for domain adaptation in CV. In NLP, words are closely linked, as in human cognition, where one word can be associated with many words. Transformers can relate features in the source and target domains. Therefore, a multi-transformer (MT) is introduced to solve the cross-domain few-shot RC by mapping features to link the features of the target and source domains and assigning different weights to the features of the target domain. Intuitively, if the features in the target domain contribute more when fused with the source domain features, then the target domain features should be assigned more weight. The acquired features are more closely associated with the source domain. Finally, the features of the source and target domains are aligned by fine-tuning only a mapping layer using the mapping mechanism of the transformer.

Therefore, multi-transformer based on prototypical enhancement network (MTPEnet) for few-shot relation classification with domain adaptation is proposed in this study. The main contributions are summarized as follows:

- The local and global representations module is designed. After obtaining embeddings with the pre-trained language model BERT, the module not only considers the entity pair of the sentence but also considers the entire sentence and obtains a better prototype with the relation representation.
- The multi-transformer is designed to align the features of the source domain and the target domain by using the mapping of the transformer and using cross-attention to assign different weights to the features in the target domain to improve accuracy in few-shot RC.
- Comprehensive experiments are conducted on the two datasets, including FewRel 2.0 and FewTAC, to clearly demonstrate the effectiveness of the MTPEnet.

The rest of this paper is structured as follows. Section 2 presents the related work. Section 3 presents the proposed MTPEnet. Section 4 presents the experimental results and analysis. Finally, Section 5 presents the conclusions of this work.

2. Related work

2.1. Few-shot relation classification

Few-shot RC trains the model on the base classes with enough data and identifies new classes with limited test data, and the purpose is to address the long-tail relation problem. It is difficult for the model to match the data distribution of the new class due to the limited labeled data. Researchers have endeavored to learn high-quality features and identify new relations with a few data. The main approaches can be classified into three categories: model fine-tuning, meta-learning, and metric learning. Early studies used transfer learning techniques to improve the model [23]. Metric learning has been proposed to learn the projection of the relation with similar relations being closer together [24–26]. Meta-learning has been proposed to improve the capability of learning new knowledge quickly by leveraging existing knowledge experiences [27,28]. Prototypical networks [14] are a simple but useful metric network frequently used in few-shot learning. There are also several variants of prototypical networks [17,29–32].

Many researchers have attempted to maximize the use of limited information due to the lack of data. Xie et al. [33] added the entity description to obtain more detailed information for a better representation. Livio et al. [34] built a relation representation model that considers relation descriptions and the potential for the same entity pair to have the same semantics. Then, Yang et al. [17] added entity and relation descriptions to improve the model performance. Zhang et al. [18] introduced a knowledge graph of the relevant domain and the concept embedding of entities as additional knowledge. As more information is added, multiple methods exist for fusing it. Liu et al. [19] proposed a straightforward and efficient method for fusing relation information into a model. However, these approaches do not consider the local and global representations of a sentence. To create a more appropriate initial prototype, the local and global representations of the sentence are considered in this study.

2.2. Few-shot relation classification with domain adaptation

Domain adaptation aims to predict tasks in the target domain by using annotated data from the source domain. Yu et al. [35] proposed a new metric learning method to address the cross-domain text classification problems. Gao et al. [16] proposed a domain adaptation problem in few-shot RC, that is, whether the model can be applied to new domains when only a few instances are available. The main problem with domain adaptation is the distribution discrepancy of the data between the source and target domains. Gao et al. [16] introduced adversarial training to improve classification accuracy. The adversarial training aligns the distributions of the two domains. The feature extractor was used to confuse the domain from which the features originated. A discriminator was used to identify whether the features originated from the source or the target domain. Feature extractors and adversarial discriminators were used to enhance domain adaptation capabilities. Cong et al. [36] introduced an inductive methodology called DaFeC to enhance the domain adaptation ability of few-shot RC using unsupervised label clustering. Yin et al. [21] proposed a multichannel encoder to capture the features from an entity and context and introduced a word-level attention mechanism based on CNNs to encode the context. The mapping-based approach maps the features of the source and target domains into similar feature spaces. Tang et al. [20] introduced a feature transformation layer in which the features of the support and query instances were transformed by affine functions, respectively, and the scaling and bias terms were optimized by meta-learning. The method introduced an entity-level attention fusion mechanism to fuse additional information.

Although these methods achieve promising results in the domain adaptive problem of few-shot RC, they ignore the relation between the features in the source and target domains. They are typically mapped separately. Therefore, the cross-transformer is used for feature mapping in this paper. The cross-attention mechanism in the transformer is used to map the features of the two domains and assign different weights to the features of the target domain based on the degree of fusion of the features of the target and source domains to better align the features of the two domains. The cross-attention mechanism can be better used in domain adaptive problems because it enables the interaction of the source and target domains rather than simply separating the target or source domains using the self-attention mechanism.

3. Methodology

In this section, the MTPEnet is presented, starting with the task definition and notations. Subsequently, the prototypical networks are reviewed. Moreover, the architecture of MTPEnet is shown. Finally, the two modules, LGR and MT, are presented separately.

3.1. Task definition and notations

In the few-shot RC, each task consists of a support set and a query set. The support set contains limited labeled instances, and the query set is a collection of several unlabeled instances. Few-shot RC tasks often use N -way K -shot settings. The support set contains N relation categories, each containing K instances. The query set is also randomly sampled from the same N relation categories. An N -way K -shot task is typically considered as an episode that better imitates few-shot scenarios [24].

During the training stage (source domain), according to the setting of N -way K -shot, several meta-tasks $\mathcal{T}$ are sampled from train dataset D_{train} , in which the support set S and the query set Q are sampled, $S = (x_i^k, r_i)$ ($i = 1, 2, \dots, N, k = 1, 2, \dots, K$), and R defines the N relations for which the instances are classified, where $R = r_1, r_2, \dots, r_N$. x_i^j is j th instance of relation r_i , each instance contains head and tail entities. Moreover, query instances are chosen at random from the rest

Table 2

Symbolic interpretation. Rep-S and Rep-R are abbreviations of the representation of the sentence and relation, respectively.

Symbol	Description
S_i_{loc}	The local Rep-S of the i th relation in the support set.
S_i_{glo}	The global Rep-S of the i th relation in the support set.
q_{loc}	The local Rep-S in the query set.
q_{glo}	The global Rep-S in the query set.
R_i_{loc}	The local Rep-R of the i th relation.
R_i_{glo}	The global Rep-R of the i th relation.
S_i/q	Enhanced prototype of the i th and q_{loc} .
$C_1, C_2, \dots, C_M$	A condensed representation of the source domain features.
S'_i_{loc}	Final prototype.
$q'_{i_{loc}}$	The local Rep-S of query instance after multi-transformer.
L_{CE}	Cross-entropy loss.

samples of N categories to create the query set $Q = (q_i^j, r_i)$ ($i = 1, 2, \dots, N, j = 1, 2, \dots, Z$). Where Z represents the number of instances of each relation in the query set, for each task $\mathcal{T}$, the model is trained using support set S and tested on query set Q .

The same settings as those in the training phase are used in the fine-tuning stage (target domain). The model is fine-tuned with limited labeled instances from the target domain to improve its domain adaptation ability.

The same settings as those in the training stage are used in the testing stage (target domain). The relation categories in task $\mathcal{T}$ do not intersect with those in the training phase. Furthermore, the support set has limited labeled instances, and the query set has no labels. Finally, the effectiveness of the model is evaluated based on the classification accuracy of the query set.

3.2. Preliminary on prototypical networks

Prototypical networks are a metric-based learning approach and were proposed for few-shot image classification by Snell et al. [14]. The core idea is to learn an embedding space to classify the classes by calculating the distance to a prototype for each category. For the few-shot task, a prototype for each category is calculated first. The optimization model then works by narrowing the distance between the corresponding prototype and the query instance. Given a new instance, only the distance between the instance and the prototype needs to be calculated, and the predicted class is determined by the class with the shortest distance. In the N -way K -shot classification, a prototype vector can represent each relation the c_i . In the vanilla approach, the prototype c_i is calculated as the mean value of the supported instances in each category.

$$c_i = \frac{1}{K} \sum_{j=1}^K x_i^j. \quad (1)$$

For Eq. (1), K denotes the number of instances of each relation in the support set. For any query instance q , the probability of belonging to the relation r_i can be calculated as follows:

$$p_\phi(y = r_i | q) = \frac{\exp(-d(E(q), c_i))}{\sum_{j=1}^N \exp(-d(E(q), c_j))}, \quad (2)$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance between two vectors. $E(q)$ denotes the vector of query instance q after the encoder.

3.3. Framework

MTPEnet framework is illustrated in Fig. 1. Table 2 lists the interpretation of the symbols in Fig. 1, detailed calculations will be shown in this section.

First, the query instances, support instances, and relation information are encoded using BERT [15]. Second, the prototypes $Proto_{loc}$ and

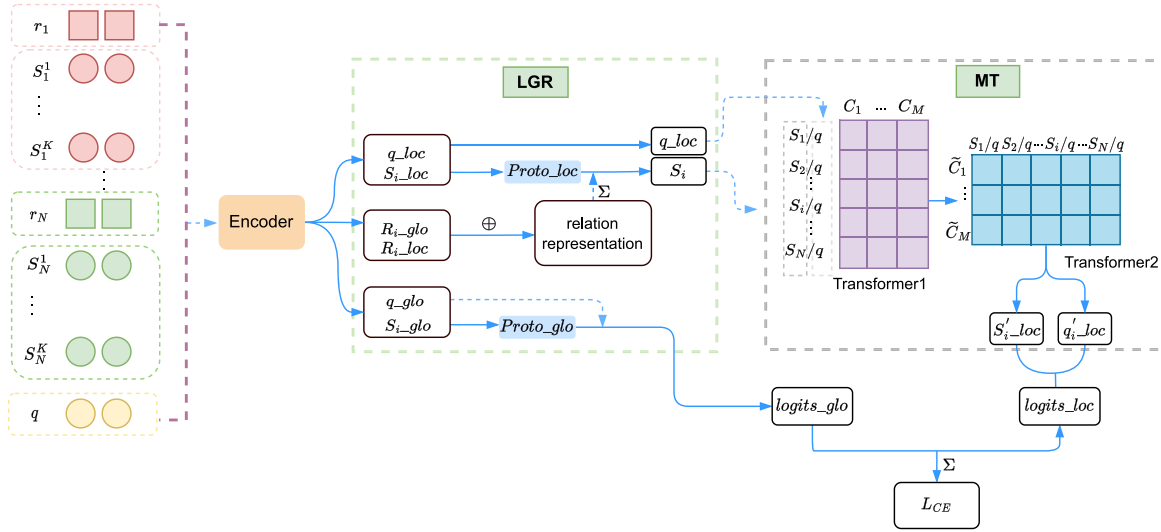


Fig. 1. Framework of the proposed MTPEnet.

$Proto_glo$ are computed according to the local and global representations of the support instances, respectively. $Proto_loc$ and the relation representation are directly added to obtain a better prototype called the enhanced prototype S_i , which is transformed together with the local representation of the query instance through the multi-transformer for feature transformation to obtain the final prototype S'_i_loc and the local representation of the query set. Third, the Euclidean distance between the final prototype of each relation class and the local representation of the query instance is calculated and represented as $logits_loc$. Subsequently, the Euclidean distance between the $Proto_glo$ represented by each relation type, and the global representation of the query instance is calculated to obtain $logits_glo$. Finally, the two logits are added together, and the cross-entropy loss is calculated using the label.

3.3.1. Local and global representations module

For a support or query instance $x = w_1, w_2, \dots, w_n$, each word w_i is represented by a word embedding v_i in the d_w dimension. According to the research idea of Zeng [1], the position embedding feature of an entity is added based on the word embedding. The position embedding dimension is d_p . After adding the position vector, the word embedding is $v_i = [w_i, pos_i^h, pos_i^t]$. Each instance can be written as an instance matrix as Eq. (3).

$$S = \{v_1, v_2, \dots, v_n\} \quad (v_i \in \mathbb{R}^{(dw+2dp)}, S \in \mathbb{R}^{(n \times (dw+2dp))}). \quad (3)$$

The BERT encoder is used to obtain the embedding of every word. After the BERT encoder, sentence embedding can be obtained. $S' = BERT(S)$.

As shown in Fig. 2, the relation interpretation and relation name for every relation are concatenated in the source domain. After BERT, two embedding representations are obtained. The “CLS” token is the hidden state at the last level of the first token of the sequence and it is not a good summary of the whole sequence [37]. The average of the embedding of all tokens considers the embedding of all words in a sentence; therefore, it is more comprehensive. The local representation of relation embedding R_loc is represented by the “CLS” token embedding and the global representation of relation embedding R_glo is represented by the average of the embeddings of all tokens. Eq. (4) is the representation for the final relation, where $\oplus$ represents the concatenation operation.

$$R^{final} = R_loc \oplus R_glo. \quad (4)$$

Moreover, for a support or query instance, the embedding representation of the entity pair and the entire sentence can be obtained using BERT. Specifically, the head and tail entities of the hidden state

of the output of the BERT encoder are concatenated for use as the local representation of the sentence. The global representation of a sentence is the average embedding of all tokens. To support instance representation, the representation of an entire sentence is considered in this study. Therefore, the proposed method is suitable for the human recognition process. When people classify a relation, they would first observe the relation name, and then understand the relation to correctly classify it. First, the labeled head and tail entities are viewed. If unfamiliar, the entire sentence is observed, and the relation is correctly classified. Observing the entire sentence is a critical step. The relation may not be correctly confirmed by knowing only the information of the entity pair, because of the possibility of diverse relations between the same head and tail entities. The entity pair is embedded in the support instance as S_loc and the embedding considers the entire sentence as S_glo . Thus, the query instance can obtain q_glo and q_loc . Then the two initial prototypes of class i can be calculated using Eq. (5), where K represents the number of instances of each relation.

$$Proto_loc = \frac{1}{K} \sum_{j=1}^K S_i^j_loc, \quad (5)$$

$$Proto_glo = \frac{1}{K} \sum_{j=1}^K S_i^j_glo.$$

The relation information is fused to obtain a vector that better represents relation i . The enhanced prototype of the support set can be expressed using Eq. (6).

$$S_i = Proto_loc + R^{final}. \quad (6)$$

3.3.2. Multi-transformer feature transformation module

A multi-transformer feature transformation module is inserted before the prototypical networks to improve the domain adaptation performance of the few-shot RC. Because of the inconsistent feature distribution between the source and target domains, the MT module takes advantage of the multi-transformer to map the various features of the source and target domains to similar feature subspaces and assigns different weights to the features of the target domain.

The MT module consists of two transformers [38], where the use of a cross-attention mechanism helps the transformer to find helpful information in the features of the source and target domains. $C = C_1, C_2, \dots, C_M$ represents the concise representation of the source domain features called the prototype*, which differs from the prototype in the prototypical networks. M denotes the number of prototypes*. Note that the prototype* represents a condensed representation of the source

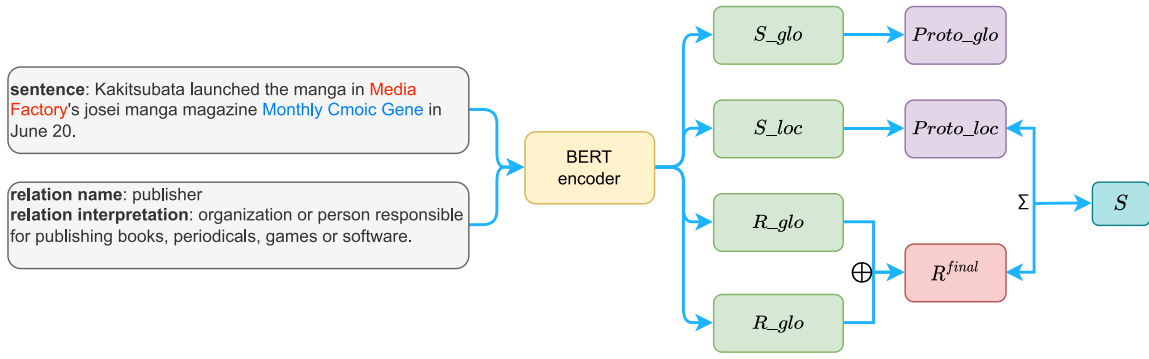


Fig. 2. Detail of LGR module.

domain for interaction with the target domain features and the prototype in the prototypical networks represents classes of relations in the embedding space. S_i/q is the abbreviation for an enhanced prototype of the i th relation class and q_{loc} . S_i can be obtained by using Eq. (6). q_{loc} denotes a local representation of the query instance after the BERT encoder. The MT module uses the S_i/q embedding representation as the input feature. The enhanced prototypes of N relations have been calculated in the LGR module, and for convenience of understanding, the i th relation is used as an example in this study. In transformer1, the input feature and prototype* obtain the intermediate features $\tilde{C} = \tilde{C}_1, \tilde{C}_2, \dots, \tilde{C}_M$ through a cross-attention mechanism. The intermediate features $\tilde{C}$ and input features S_i/q go through transformer2 to obtain the features that finally enter the prototypical networks. Because the initial input is S_i/q , the features adapted after the module are S'_i/q' .

$$\tilde{C} = \text{Transformer1}(S_i/q, C), \quad (7)$$

$$S'_i/q' = \text{Transformer2}(\tilde{C}, S_i/q) + S_i/q. \quad (8)$$

In Eq. (8), the original input features and the transformed features are connected by residuals to obtain the final features. For each transformer, the dimensions of the attention matrix are $N \times M$. Specifically, the number of feature points $N > 10^4$, and $M = 256$ is selected. This implies that the source domain is simplified to a set of 256 prototypes*.

Take transformer1 as an example of a multi-transformer:

$$\text{Att_weight}(S_i/q, C) = \sigma(K(S_i/q), Q(C)) \in R^{N \times M}, \quad (9)$$

$$\text{Attention}(S_i/q, C) = \text{Att_weight}(S_i/q, C) \cdot V(S_i/q), \quad (10)$$

$$\tilde{C} = \text{FFN}(\text{Attention}(S_i/q, C)). \quad (11)$$

$\text{Att_weight}(S_i/q, C)$ represents the degree of contribution of the input features to the fusion of source domain features. The larger the weight of the feature, the better it can be fused with the source domain features. Finally, the more suitable source domain features would be obtained, and the target domain features would be better aligned with the source domain features. FFN represents a feed-forward network, which mainly adds non-linear transformations to make the word representation more powerful. The fused features are then converted into $\tilde{C}$. In Eq. (9), K controls which subspace the input feature S_i/q maps. Furthermore, it can affect the attention between the input features and the prototype*. According to the assumption in [22], in the target domain, by updating K , the input feature can be mapped to a feature subspace similar to the source domain. Then, when training a new domain in the target domain, Q and V can be shared with the source domain, and only K needs to be updated; thus, more information about the source domain is retained.

3.3.3. loss function

The local and global representations of the sentence are also considered when the loss is calculated. An enhanced prototype S_i and the local representations of the query instances q are feature transformed by the multi-transformer to obtain the final prototype and final local representation of the query instances. The total logits can be calculated.

$$\text{logits} = \text{logits}_{loc} + \text{logits}_{glo}, \quad (12)$$

where logits_{loc} can be computed based on the final prototype S'_i_{loc} and final local representation query instance q'_{loc} . The $Proto_{glo}$, which is computed using the global representation of the support instance, computes logits_{loc} together with the global representation of the query instance. Because the global representation dimension of a sentence is already high, it no longer passes through multi-transformer feature transformation module. The coefficients are set to 1 in Eq. (12), because equal importance is considered both global and local representations. The cross-entropy (CE) loss function is used in this paper to calculate the loss. The query instances are categorized by using a distance metric function. Then, the final loss can be expressed as:

$$\text{loss} = \text{CE}(\text{logits}, \text{label}), \quad (13)$$

where logits represents the Euclidean distance from each query instance to the prototype of each relation class, considering the local and global representations. label represents the actual label of a sentence.

4. Experiments and comparative analysis

In this section, to evaluate the MTPEnet, experiments are conducted on two datasets FewRel 2.0 and FewTAC. Moreover, two ablation studies verify the performance of the proposed LGR and MT modules.

4.1. Dataset and evaluation metrics

MTPEnet is evaluated using FewRel 2.0 [16] and FewTAC [21]. FewRel 2.0 is a benchmark dataset, where the training set is from Wikipedia, and the testing set is from PubMed. The dataset FewRel 2.0 has 74 relations, each with 700 instances. Because only the 10 validation tasks and 64 training tasks are publicly available, the experimental results of the FewRel 2.0 in this study are obtained from the 10 validation tasks. Moreover, the fine-tuning of MTPEnet is performed by using SemEval-2010 task 8 datasets, because there are no more data available for the PubMed field. As the fine-tuning dataset had only 17 relation categories, the 10-way 1-shot, and 10-way 5-shot settings are not satisfied during training and testing in the target domain. Because training and evaluating the model requires ten categories, three additional relations from the NYT dataset are added. During fine-tuning, both the training and validation tasks consist of ten relations. No overlapping data exists among the training, validation, or testing datasets.

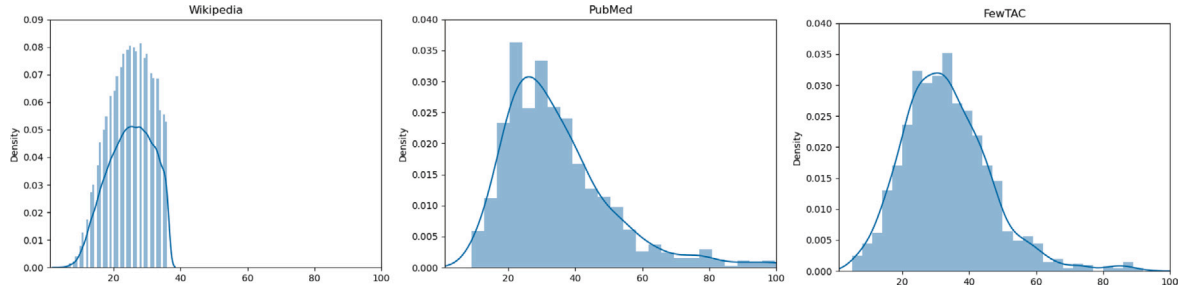


Fig. 3. Sentence length distributions of Wikipedia, PubMed, and FewTAC.

Table 3

Settings of datasets. R_N and I_N mean the number of relation types and the total number of instances. S and T represent data from the source and target domains respectively. FT indicates fine-tuning.

Dataset	Domain	Task	R_N	I_N
FewRel 2.0	S-Wikipedia	Training	64	44,800
	S-Wikipedia	Validation	16	11,200
	T-SemEval	Ft-training	10	4364
	T-SemEval+NYT	Ft-validation	10	4787
	T-PubMed	Test	10	1000
FewTAC	S-Wikipedia	Training	64	44,800
	S-Wikipedia	Validation	16	11,200
	T-Re-TAC	Ft-training	11	3723
	T-Re-TAC	Ft-validation	10	1726
	T-Re-TAC	Test	21	9551

Table 4

Parameter setting.

Parameter name	Value
Maximum instance length L	128
Word embedding size d_w	768
Hidden layer dimension d_h	230
Learning rate	-1
Batch size	1
Weight decay	10^{-5}
Optimizer	SGD

The dataset FewTAC is also used to validate the model. It has the same training data as FewRel 2.0. The test dataset is from ReTACRED [39], which is from the news domain, and it is very different from the Wikipedia sentences. The FewTAC dataset contains a total of 41 relations, and this dataset is split for fine-tuning, with the remaining 21 relations used for testing, except for the necessary 20 relations. The settings for each step are listed in Table 3. The sentence lengths for the three domains are shown in Fig. 3. It is worth noting that the sentence length distributions are very different, which makes it difficult to solve domain adaptation problem in few-shot RC.

4.2. Implementation details

The episodic training strategy is adopted in this study. Each episode is an N -way K -shot task. For dataset FewRel 2.0, as a common setting, N is set to 5 or 10, and K is set to 1 or 5 as in Han et al. [10]. N relations are randomly selected to construct a training task relation set, and K instances from each relation are used to construct a support set. Q instances are randomly selected from the above N classes as the query set. For the dataset FewTAC, K is set to 5 in the experiments, following by Yin et al. [21]. MTPEnet is trained and evaluated on two NVIDIA A5000 GPUs (24 GB). Table 4 lists the various hyperparameters. The encoder BERT is used to generate the word embeddings of the sentence and relation. The model is trained for 30,000 epochs and evaluated every 1000 epochs. All the models are trained on the training task, and the best epochs are chosen to obtain the test results. The

detailed process of training in the source domain and adaptation to the target domain is as follows:

Source domain: First, the prototypical networks are trained in the source domain. Second, the parameters of the prototypical networks are frozen, and the weights of the multi-transformer are trained in the source domain. The same training data are available in the source domain. Finally, after training, the prototypes* are a condensed representation of the source domain features in a multi-transformer.

Target domain: First, the parameters are fine-tuned in the target domain, and only the K projection weights are updated. The results are then obtained by predicting the relation of the query instances in the testing set.

4.3. Competitive baselines

Prototypical networks have the advantages of simplicity and effectiveness, so it is used as the few-shot classifier in this paper. To corroborate the superiority of the model, some representative prototypical-based network models for the few-shot RC task are chosen. Because MTPEnet does not introduce the domain knowledge base, it is not compared with the model that introduces an additional knowledge [18, 40].

Classical prototypical networks are denoted as Proto. Many models improve on it for few-shot RC, including Proto (CNN) [10], Proto (BERT) [16], Proto-ADV(CNN) [10], Proto-ADV(BERT) [10], DaFeC (BERT) [36], BERT-PAIR [16], KDFT(BERT) [20], Proto-MCE+SR [21]. CNN and BERT represent different encoders for sentence encoding. “-ADV” represents the use of adversarial learning in the model.

Proto (CNN/BERT) are metric-based classical prototypical networks. The base prototype of each relation is calculated by averaging the embedding of the support instance.

Proto-ADV (CNN/BERT) add adversarial training to the prototypical networks.

DaFeC (BERT) is an unsupervised domain adaptation framework using clustering to improve the performance of few-shot RC. First, the feature extraction method groups unlabeled data in the target domain. The source domain data and generated pseudo-label data are utilized as supervision to update the parameters of the few-shot classifier.

BERT-PAIR is a sequence classification algorithm in BERT. Each instance of a query is paired with every instance of support, and each pair is concatenated as a sequence. The connection sequence is fed into the BERT sequence classification algorithm to obtain scores, and the sentence with the highest score in the support set is selected. The relation predicted by the query set is the relation of the sentences that obtain the highest scores.

KDFT aims at the domain adaptation problem in few-shot RC. A diverse feature transformation layer is inserted into the encoder to learn a broader range of features and enhance the effect of domain adaptation.

Proto-MCE+SR improves the performance of CNNs-based prototypical networks in few-shot RC. A multichannel instance encoder (MCE) is proposed to capture domain-variant features. A Selection strategy (SR) is used to select high-confidence instances.

Table 5
Accuracies (%) of different models on datasets FewRel 2.0 and FewTAC.

Model	FewRel 2.0				FewTAC	
	5-way 1-shot	5-way 5-shot	10-way 1-shot	10-way 5-shot	5-way 5-shot	10-way 5-shot
Proto (CNN)	35.09	49.37	22.98	35.22	73.83	60.36
Proto-ADV (CNN)	42.21	58.71	28.91	44.35	74.21	61.06
Proto (BERT)	40.12	51.50	26.45	36.93	81.42	68.07
Proto-ADV (BERT)	41.90	54.74	27.36	37.40	82.31	70.62
DaFeC (BERT)	46.39	56.32	32.09	40.53	83.81	72.28
BERT-PAIR (BERT)	56.25	67.44	43.64	53.17	84.01	73.13
KDFT (BERT)	58.01	81.92	46.24	59.42	–	–
Proto-MCE+SR (CNN)	–	76.12	–	65.34	85.32	74.11
MTPEnet (this work)	64.67	88.87	57.86	80.19	91.76	84.50

Table 6
Accuracy (%) of ablation study on FewRel 2.0 validation set.

Model	5-way 1-shot	5-way 5-shot	10-way 1-shot	10-way 5-shot
Baseline	40.12	51.5	26.45	36.93
+ fine-tuning	46.48	53.51	29.55	48.33
+ fine-tuning + MT	55.61	76.68	36.44	62.66
+ fine-tuning + MT + LGR	64.67	88.87	57.86	80.19

4.4. Comparative results and analysis

Table 5 shows the results of all the compared models for FewRel 2.0 and FewTAC. It can be observed that MTPEnet outperforms all the benchmark models. MTPEnet is based on the Proto (BERT) method, which does not introduce the improved modules.

From Table 5, the MTPEnet's classification accuracy is higher than that of the basic model Proto (BERT) with large margins of 24.55%, 37.37%, 31.41%, and 43.26% for the four few-shot settings respectively in FewRel 2.0. The change in accuracy is visualized in Fig. 4, which confirms MTPEnet has good effectiveness in cross-domain few-shot RC tasks. The accuracy improved by 10.34% and 16.43% on 5-way 5-shot and 10-way 5-shot tasks respectively, in FewTAC. The figure above demonstrates that the inconsistent feature distribution problem can be alleviated by the cross-attention mechanism, and the source domain and target domain data are then mapped to similar subspaces by the transformer. MTPEnet improves the classification capabilities of the trained model for the cross-domain case of few-shot RC. Simultaneously, a better semantic representation can be obtained, and the performance of the model can be further enhanced by combining the detailed information of the name and the description of the relation, considering the representation of the sentence from the global and local perspectives. Because there is no detailed relation description of the target domain in the datasets FewRel 2.0 and FewTAC, only the semantic representation of the relation name in the target domain fine-tuning and the model can still obtain better classification ability.

KDFT was only tested on FewRel 2.0, and Table 5 lists the results provided in the original paper. The results for the dataset FewTAC are not available in Table 5 because no available code is provided. KDFT combines the entity information and improves semantic intersection, and different feature transformation layers are used to improve the domain adaptation effect. Nevertheless, it ignores information from the entire sentence and comprehensive semantic representations, which are vital for understanding and classifying relations. Although both feature transformation layers are used, as with KDFT, MTPEnet performs significantly better with two transformers than with an affine transformation. In particular, in the 10-way 5-shot task on the dataset FewRel 2.0, MTPEnet is 21% higher than KDFT, indicating that the MT and LGR modules can significantly improve the effectiveness of this task. Proto-MCE+SR used an attention mechanism based on CNNs in order to capture the domain-invariant features. However, the CNNs-based model may not extract semantic information accurately when there is limited data. Table 5 lacks the results of the 1-shot task because it is not published in their paper.

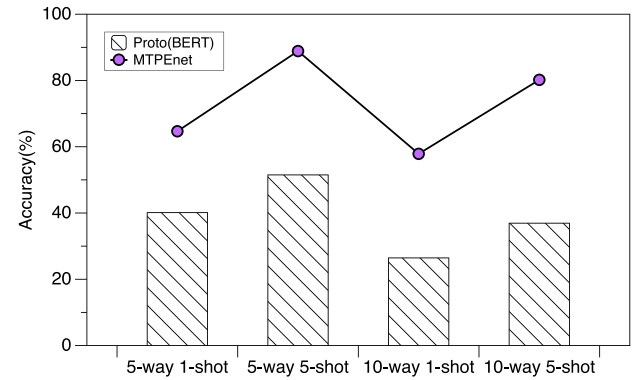


Fig. 4. Comparison of accuracy in dataset FewRel 2.0 for Proto (BERT) and MTPEnet.

4.5. Ablation study

Ablation experiments were conducted to demonstrate the effectiveness of each module.

4.5.1. Effect of local and global representations

The goal of LGR is to provide a better semantic representation of relations and sentences. The results of the ablation experiment of the LGR module are shown in row five of Table 6, demonstrating that considering local and global representations while adding relevant information of relations (relation names and relation interpretations) can enhance the accuracy of the trained model for the classification of few-shot RC. Table 6 (last row) shows that the model improves twice on the 10-way task compared with the 5-way task. As it becomes more difficult for the model to select the correct relation in the few-shot RC task when the number of relations increases, it is important to consider a better semantic representation. MTPEnet considers the local and global representations, both the entity pair and the interaction of entities between the entire sentence, and performs the semantic representation of relations and sentences. Experiments are conducted to compare the classification effects of considering only local and considering the entity pair and the entire sentence in the source and target domain. The experimental results for the source domain and target domains are presented in Tables 7 and Table 8. From Table 7, the model is slightly lower than the previous model, considering only the local representation at 5-shot tasks in the source domain. This is because, in the 5-shot task, there are more instances of each relation

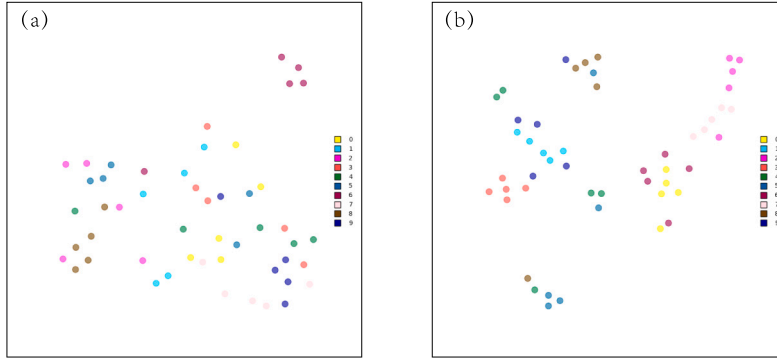


Fig. 5. T-SNE feature visualization. (a) Without multi-transformer. (b) With multi-transformer.

Table 7

In source domains, consider parts of the sentence and consider all.

Setting	Consider part	Consider all
5-way 1-shot	89.15	89.45↑
5-way 5-shot	94.82	94.20↓
10-way 1-shot	81.64	83.36↑
10-way 5-shot	91.94	91.36↓
Average	89.39	90.34↑

Table 8

In target domains, consider parts of the sentence and consider all.

Setting	Consider part	Consider all
5-way 1-shot	43.38	44.74↑
5-way 5-shot	67.15	71.81↑
10-way 1-shot	29.70	30.83↑
10-way 5-shot	59.27	63.53↑
Average	49.87	52.72↑

and there may be a more comprehensive semantic representation already available. However, in the 1-shot task, considering the sentence comprehensively is better than considering a partial representation of the sentence. Finally, the overall classification is higher in the four settings, indicating that considering the entire sentence is more suitable for few-shot models. Moreover, as shown in Table 8, in the cross-domain few-shot RC, the dataset contains only the name of the relation but not a specific description of the relation. Therefore, only the name of the relation has been added. Global and local representation is better than considering only the entity pair for all four tasks. This indicates that the approach performs better for cross-domain tasks. Moreover, this might be because the training model is only trained on the source domain, and there is very little data in the target domain. Therefore, it becomes particularly important to fully consider the information in the sentences fully.

4.5.2. Effect of multi-transformer

The goal of MT is to reduce the large discrepancy in the feature distribution between the source and target domains. The results of the ablation experiments of the MT module are presented in row four of Table 6, indicating that, by mapping the features of the source and target domains to similar feature subspaces and using the advantages of the transformer, the MT can increase the classification accuracy of the model. However, MTPEnet is fine-tuned in the target domain, that is, the fine-tuning dataset from a different corpus than the validation dataset, but in the same field. Relevant experiments were conducted to eliminate the effect of the fine-tuning dataset. Row three and four of Table 6 indicate that the classification accuracy of the model is greatly improved by adding the MT module compared with the method that is only fine-tuned in the target domain, verifying the effectiveness of

the MT. In addition, compared with the baseline model, adding fine-tuning and MT can improve the accuracy by approximately 15% and 25% for the tasks of 1-shot and 5-shot. When using the cross-attention mechanism, the more instances of each class, the more accurate the weights of the features of the target domain obtained. Moreover, for 5-shot tasks, the effect of fine-tuning is weaker than that of MT, with an average increase of only 7%. That is to say. The MT achieved more remarkable improvement in the 5-shot task. Visualized feature distribution maps of the sentence embedding for the dataset FewRel 2.0 are obtained in Fig. 5 by the t-distributed stochastic neighbor embedding (T-SNE) technique. Testing the 10-way 5-shot task in the target domain, the left panel represents the distribution of features without the MT module and the right panel with the MT module. The figure indicates that the sentence embeddings of the same relation are closer after adding the MT module.

5. Conclusions

In this study, MTPEnet is used for few-shot relation classification with domain adaptation. First, the local and global representations module is proposed. The local and global information of a sentence is considered during encoding. Thus a better prototype representation can be obtained using prototypical networks. The experimental results show that the proposed method improves the overall accuracy, particularly in cross-domain tasks. Second, a multi-transformer feature transformation module is used to solve the domain adaptation problem. The features of the source and target domains are mapped to similar feature subspaces by using transformer mapping. Moreover, the cross-attention mechanism of the transformer allows the features in the target domains to be assigned different weights. Comprehensive experiments, including ablation experiments on two datasets, demonstrated the effectiveness of the proposed method.

CRedit authorship contribution statement

Hongfei Du: Conceptualization, Methodology, Data curation, Validation, Writing – original draft, Writing – review & editing. **Qinghua Zhang:** Supervision, Funding acquisition, Project administration, Resources, Writing – review & editing. **Keyuan Li:** Data curation, Investigation. **Fan Zhao:** Writing – review & editing, Formal analysis. **Guoyin Wang:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Hongfei Du received the B.S. from the Sichuan University of Science and Engineering, Sichuan, China, in 2021, where she is currently pursuing the M.S. degree. Her research interests include natural language processing and artificial intelligence.



Qinghua Zhang received the B.S. degree from Sichuan University, Chengdu, China, in 1998, the M.S. degree from the Chongqing University of Posts and Telecommunications, Chongqing, China, in 2003, and the Ph.D. degree from Southwest Jiaotong University, Chengdu, in 2010. He was at San Jose State University, San Jose, CA, USA, as a Visiting Scholar in 2015. Since 1998, he has been at the Chongqing University of Posts and Telecommunications, where he is currently a Professor and the Director of the Science and Technology Division. His research interests include rough sets, fuzzy sets, granular computing, and uncertain information processing.



Keyuan Li received the B.S. degree in the School of Science from Chongqing University of Posts and Telecommunications in 2021. Currently, he is working toward the M.S. degree in the School of Computer Science and Technology, Chongqing University of Posts and Telecommunications. His research interests include uncertain information processing, broad learning system, ensemble learning, medical image processing.



Fan Zhao received the B.S. and M.S. degrees from the Chongqing University of Posts and Telecommunications, Chongqing, China, in 2017 and 2020, respectively, where she is currently pursuing the Ph.D. degree. Her research interests include uncertain information processing, three-way decisions, fuzzy sets, granular computing, and rough sets.



Guoyin Wang received the B.S., M.S., and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1992, 1994, and 1996, respectively. From 1998 to 1999, he was at the University of North Texas, Denton, TX, USA, and the University of Regina, Regina, SK, Canada, as a Visiting Scholar. Since 1996, he has been at the Chongqing University of Posts and Telecommunications, Chongqing, China, where he is currently a Professor, the Director of the Chongqing Key Laboratory of Computational Intelligence, and the Director of the Key Laboratory of Tourism Multi-source Data Perception and Decision, Ministry of Culture and Tourism. He was appointed as the Vice-President of the Chongqing University of Posts and Telecommunications. He has more than 200 reviewed research publications. His research interests include rough sets, granular computing, knowledge technology, data mining, neural networks, and cognitive computing.